Lexical Split Design and Representation-Interface Validity in Decoder-Only Morphology Probing: An Arabic Evaluation Study

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Abstract

Diagnostic classifiers are often evaluated with random instance splits even when lexical identity is correlated with the target label. This paper treats split design and representation-interface design as measurement choices in decoder-only morphology probing. Using Arabic morphology as a testbed, we evaluate eleven pinned Hugging Face models with BF16 hidden-state captures, three morphology tasks, three split conditions, and four representation interfaces. The evaluation uses deterministic nested five-fold procedures, development-only layer selection, fixed Ridge classification, shuffled-label controls, validated token alignment, and checksummed artifacts. Across the 132 model–task– interface comparisons, random-split macro-F1 is higher than both lemma-heldout and root-heldout macro-F1. The random-minus-lemma differences range from 0.0127 to 0.2863 (mean 0.1059), and random-minus-root differences range from 0.0255 to 0.2796 (mean 0.1179). Effects vary by model, task, and interface; the range rules out a single summary penalty. A full-metadata prompt-final state compares position together with information availability: the prompt explicitly supplies lemma, root, and pattern. A metadata-free prompt-final condition and target-span pooling expose these confounds while holding the dataset and probe fixed. The results support claims about linear recoverability under named prompts, splits, token spans, and estimators; questions about causal model use, linguistic competence, and behavior in other settings require separate evidence. Planned post-extraction analyses can use archived representations and derived artifacts alone.

Keywords: Probing, lexical leakage, split design, Arabic morphology, decoder-only models, reproducibility

1 Introduction

Diagnostic classifiers are useful because they make a representation's accessibility testable with a comparatively simple estimator. A probe that predicts a linguistic label from a hidden state requires an explicit account of its measurement conditions. The result can depend on which examples are placed together, which information the prompt supplies, where the hidden state is extracted, how subword tokens are aligned to the stimulus, and how imbalanced labels are scored. Work on probing has therefore emphasized control tasks, selectivity, and careful interpretation (Hewitt and Liang 2019; Hewitt and Manning 2019; Tenney et al. 2019b; Ravichander et al. 2021).

This paper studies one particularly consequential evaluation choice: whether the test set contains lexical identities or lexical families that also occur in training. Under a random instance split, repeated lemmas, roots, or related surface forms can appear on both sides of the partition. If those identities are correlated with morphology, a probe can obtain a high score while being tested mainly on interpolation within a familiar lexical inventory. A split that holds out lemma or root groups asks a different question: how much of the label remains linearly recoverable when the grouping identity is absent from training? Random splitting serves as a legitimate optimistic reference for instance-level interpolation; its interpretation remains distinct from lexical generalization across heldout identities. The distinction follows the broader warning that standard and random splits can be misaligned with the intended generalization claim (Gorman and Bedrick 2019; Søgaard et al. 2021).

Arabic provides a useful testbed because lexical, templatic, and inflectional structure interact. Lemmas and roots define meaningful grouping variables, yet holding them out leaves semantic, orthographic, and morphological relations available across partitions. The setting therefore supports a direct evaluation of lexical split sensitivity while leaving the broader open-set language question unresolved. Arabic resources also make the tokenization problem concrete: a morphological word may span multiple subword units, and different tokenizers segment the same stimulus in different ways (Taji et al. 2017; Sennrich et al. 2016; Bostrom and Durrett 2020).

Split design is only one interface choice. A decoder-only prompt-final state has processed the target, the instruction, and any fields that follow the target. In the full-metadata condition used here, lemma, root, and pattern are explicitly present. A target-final state is extracted from a causal prefix ending at the final target-overlapping token, excluding those later fields. A comparison between these states therefore changes both position and available information. We add a metadata-free prompt-final condition to isolate part of that confound and retain the complete target span so that final-subtoken and mean-span representations can be derived locally.

The central contribution is a methodological evaluation of lexical split validity and representation-interface validity in decoder-only morphology probing, with Arabic morphology as the testbed. The study uses the completed BF16 matrix, with the historical Q8 run serving as robustness context. Its contribution is an audited account of the decisions that determine what a probe score means, together with archived representations that make later analyses independent of renewed model inference.

We ask four questions:

1. **Split validity:** How much do random-instance macro-F1 estimates differ from lemma-heldout and root-heldout estimates under the same task, model, and representation interface?
2. **Interface validity:** How do full-metadata prompt-final, metadata-free prompt-final, final target-subtoken, and mean target-span interfaces differ when the information available at each site is stated explicitly?
3. **Model heterogeneity:** Are split effects stable enough to report as one penalty, or do they vary across model families, sizes, and morphology tasks?
4. **Reproducibility:** Can complete, validated BF16 captures support the planned local analyses using archived captures alone?

The principal findings are descriptive. Random-split macro-F1 exceeds both grouped heldout conditions in every row of the generated 132-row comparison table, but the differences range substantially. The full-metadata prompt-final interface is often the strongest absolute condition in aggregate, while the metadata-free and target interfaces retain substantial signal in many cells. The exact ordering changes across tasks and models. These patterns support a methodological recommendation: probing reports should state the lexical unit held out, the information available at the extraction site, the pooling rule, and the estimator-selection boundary. Linear probe performance should be interpreted as recoverability under those conditions, with causal use and linguistic competence treated as separate questions.

2 Background and Related Work

2.1 Probing and diagnostic classifiers

Probing studies use a classifier or structured estimator to ask whether a property is accessible from a representation. Structural probes, layer-wise diagnostics, and contextual analyses have made this approach a productive way to inspect learned representations (Hewitt and Manning 2019; Tenney et al. 2019a,b). The appeal of a linear probe is also its limitation: a successful classifier establishes a property of the representation–estimator pair. Its relationship to the model's generation mechanism requires separate evidence. A linear classifier can exploit distributional regularities that are unrelated to the model's causal computation for a task.

Probe capacity and controls matter. A high-capacity probe can recover information that is present in a distributed or incidental form, and a low score can reflect a mismatch between the estimator and the representation, leaving property recoverability unresolved. Control tasks and shuffled labels are therefore useful sanity checks, while selectivity and task relevance require separate reasoning (Hewitt and Liang 2019; Ravichander et al. 2021). We use a fixed standardized Ridge classifier as a deliberately limited, reproducible measurement. Our claims concern linear recoverability under that estimator; other probe classes constitute separate measurements.

2.2 Lexical memorization and split design

The random split convention is attractive because it is simple and often produces balanced partitions. It can nevertheless give an optimistic estimate when examples are clustered by a lexical or document identity. The same split-design issue arises across empirical tasks: correlated observations can be distributed across training and test while the intended generalization unit is obscured (Gorman and Bedrick 2019; Søgaard et al. 2021). In a morphology resource, lemma and root are natural grouping candidates because they capture recurrent lexical structure. The choice is substantive: a lemma-heldout score asks about transfer across lemmas, while a root-heldout score asks about transfer across a broader and linguistically different group.

Our design treats random performance as a reference condition and measures its gap from two grouped conditions. The gap is evidence that evaluation depends on lexical composition; it measures split sensitivity, while the source of that sensitivity remains distributed across several possible leakage paths. Related surface forms, morphology patterns, meanings, source-document regularities, and tokenizer fragments may cross a lemma or root boundary.

2.3 Morphology probing and Arabic resources

Morphological labels are attractive probing targets because they are structured and can be attached to individual word forms. They are also heterogeneous: part of speech, gender, and number differ in class cardinality, imbalance, and the extent to which their cues are tied to lexical identity. Accuracy can hide poor minority-class behavior, especially for three-way number classification; macro-F1 is consequently reported alongside accuracy and the training-only majority baseline.

Arabic adds both linguistic and resource-specific issues. Universal Dependencies Arabic-PADT provides a documented dependency-treebank resource for Arabic (Taji et al. 2017); CAMeL Tools provides open-source Arabic processing infrastructure (Obeid et al. 2020). Morphological analyzers can supply rich features but also introduce analyzer-specific conventions, ambiguity decisions, and normalization choices. This paper treats the resulting annotations as a constructed evaluation resource with recorded field lineage and bounded coverage of Arabic morphology. Recent Arabic analyzers illustrate both the value and the continuing engineering requirements of such resources (Khairallah et al. 2024).

2.4 Decoder-only extraction, tokenization, and pooling

Decoder-only models expose a sequence of causal hidden states along the prompt, and word representations vary with the extraction location. The state at the end of a prompt can summarize fields that follow the target; the target state excludes those later fields. Subword tokenization further means that a target word may occupy one, two, or more tokenizer pieces. Subword units are useful for efficient language modeling (Sennrich et al. 2016), while their boundaries can diverge from morphological boundaries, and tokenizer choices affect which state is called the "word" state (Bostrom and Durrett 2020). We therefore save every target-overlapping token state and derive final-subtoken and mean-span pooling after extraction. The pooling comparison tests robustness of the measurement interface and leaves linguistic canonicality as an open question.

2.5 Evaluation and reproducibility methodology

Reproducibility in representation probing has two layers. The statistical pipeline must prevent test information from selecting layers, preprocessing, or hyperparameters. The extraction pipeline must make model identity, tokenizer identity, prompts, target spans, tensor semantics, and file integrity inspectable. We address both layers with deterministic manifests, structural alignment checks, content hashes, and independently validated per-model bundles. This is especially important when model inference is performed on ephemeral GPU instances and subsequent scientific analysis is intended to run locally.

3 Evaluation Resource

3.1 Source resource and release identity

The authoritative input is a 4,701-row normalized JSONL artifact derived from the official Universal Dependencies Arabic-PADT repository: https://github.com/UniversalDependencies/UD_Arabic-PADT. The local record identifies the earliest matching official release as r2.17 and retains the archived commit and the SHA-256 of the training file. The original source copy lacks Git metadata, and the same training-file content is byte-identical in official r2.17 and r2.18. The repository therefore records the release ambiguity and leaves the historical local label unresolved. The upstream license is CC BY-NC-SA 3.0; the archived license file is retained with the source material. No acquisition date was recorded.

Only the PADT training file enters the construction pipeline. The archived development and test files are retained for provenance only. “Train,” “development,” and “test” in the probing evaluation refer to newly constructed folds, distinct from the original UD file partitions.

3.2 Construction and field lineage

The resource construction is deterministic and has two conceptually distinct stages. First, the pipeline reads eligible PADT tokens in source order, disambiguates complete sentences with CAMeL Tools 1.6.0 and the `calima-msa-r13` MLE model, takes the top analysis, and stops after 5,000 retained candidates. The construction consumed 330 sentence units and 11,437 source tokens before writing those candidates. Second, it normalizes the analyzer output, requires lemma, root, abstract pattern, and concrete pattern, removes ambiguous records, and retains normalized POS values NOUN, VERB, and ADJ. The strict filter removes 299 records and leaves 4,701.

The field lineage is explicit:

- `surface` is the analyzer/PADT surface form and `surface_dediac` is the normalized form produced by the recorded Arabic diacritic and formatting-mark removal routine.
- `lemma`, `root`, `abstract_pattern`, and `concrete_pattern` are analyzer-derived fields after the recorded normalization and ambiguity filtering.

Table 1 Counts by POS, gender, and number for rows with all three primary labels. The 45 stored `def` values are counted as semantic `du` for this descriptive table.

	Masc.			Fem.		
POS	sg	pl	du	sg	pl	du
NOUN	1580	388	25	712	243	8
ADJ	415	17	1	471	1	4
VERB	464	48	4	312	3	3

- `pos`, `gender`, and `number` are task labels selected from the normalized analyzer fields. POS has classes NOUN, VERB, and ADJ; gender has fem and masc; number has sg, pl, and du.
- `id` is an immutable stimulus identifier. Source identity, source split metadata, and the construction provenance string are retained as metadata for grouping and audit, separate from model inputs.

The frozen JSONL contains 1,519 unique lemmas, 707 unique roots, and 2,331 unique surface forms. POS counts are 2,957 NOUN, 910 ADJ, and 834 VERB. Gender is available for 4,699 rows: 2,942 masculine and 1,757 feminine, with two missing values. Number is available for 4,699 rows: 3,954 singular, 700 plural, and 45 historical dual labels stored as `def`, with two missing values. The historical spelling is preserved byte-for-byte. At task load, only the number field is canonicalized from `def` to the semantic label `du`; the dataset artifact remains byte-for-byte unchanged.

The cross-tabulation of the three primary labels after this task-specific canonicalization is shown in Table 1. It covers the 4,699 rows with all three labels and is included to make the class composition visible; it introduces no new evaluation target.

The field definitions describe this constructed resource with its analyzer ambiguity conventions. The probe unit is an eligible PADT token. No new clitic segmentation is introduced by the probing adapter. Patterns, roots, and lemmas are retained for grouping, metadata, and audit purposes; high-cardinality label prediction falls outside the primary task scope.

3.3 Original construction split and probing folds

For historical dataset construction, the vendored pipeline recorded a root-heldout split with seed 17 and nominal 0.70/0.15/0.15 proportions: 3,291 training, 705 development, and 705 test records. This construction split is distinct from the evaluation split used for the reported five-fold results. The latter is generated independently at task load from the frozen example order, as specified in Section 4.

3.4 Released and archived artifacts

The resource package includes the normalized stimulus file, source and transformation provenance, source-content hashes, the vendored construction pipeline, task-specific fold manifests, prompt specifications, BF16 model manifest, representation-bundle manifests, derived representations, result tables, prediction records for the primary models, validation reports, and checksums. Large hidden-state arrays are recorded as external artifacts in the BF16 freeze; their hashes and bundle identities are preserved, while any public redistribution must respect the relevant model licenses. The scientific evidence package contains representation artifacts and metadata; model weights remain external.

4 Evaluation Methodology

4.1 Evaluation questions and split conditions

Each task is evaluated under three split conditions. The random condition uses nested five-fold stratified partitioning with seed 42. It is the optimistic reference for ordinary instance-level interpolation. The lemma-heldout condition uses nested five-fold GroupKFold with lemma as the grouping key. The root-heldout condition uses the same procedure with root as the grouping key. For each outer fold, the test partition is the selected outer fold and the development partition is fold zero of a second partition over the outer-training data. The remaining data are training data.

The grouping conditions answer different questions. Lemma holdout keeps each lemma identity within a single partition; root holdout applies the same rule to each root identity. Direct set-intersection audits verify zero overlap of the applicable grouping key across training, development, and test in every recorded fold. These controls reduce direct lexical recurrence while leaving semantic, orthographic, morphological, and source-document correlations available across partitions. Random splitting remains the labeled reference condition for instance-level interpolation.

Rows with a missing task label are removed before fold construction. Thus POS uses all 4,701 records and gender and number use 4,699 records. The same frozen example order and deterministic split generator are used across representation interfaces for a given model and task.

4.2 Probe, layer selection, and metrics

The primary estimator is a pipeline consisting of a training-fitted `StandardScaler` followed by `RidgeClassifier` with fixed $\alpha = 1.0$. For every outer fold and every layer, the scaler and probe are fit on training examples only. Development accuracy selects the layer; development macro-F1 is the first tie-breaker, followed by the lowest layer index. The chosen layer is refit on training plus development examples and is evaluated once on the outer test partition. Test labels and test metrics never select a layer, regularization value, preprocessing operation, or model.

Accuracy and macro-F1 are both reported. Accuracy makes the majority control transparent; macro-F1 weights classes equally and exposes the effect of poor minority-class prediction. Across the five outer folds, the primary point estimate is the arithmetic mean and the reported fold dispersion is the population standard deviation across those five recorded test results. This dispersion describes fold variation; the bootstrap intervals reported elsewhere have a separate interpretation.

4.3 Baselines and shuffled-label controls

The majority baseline predicts the most frequent training label, with explicit deterministic tie-breaking. It is computed separately for every task and outer fold, never from the test labels. Shuffled-label controls are run for the two primary models, all four interfaces, all three tasks, all three split conditions, and all five folds. Labels are permuted independently within training, development, and test before the same layer-selection and fitting pipeline is run. The complete audit contains 360 shuffled cells. None exceeds its corresponding train-only majority baseline. After the label relationship is removed, the shuffled pipeline stays at the majority reference; probe performance on the real labels remains evidence about estimator-accessible information under the stated interface.

4.4 Uncertainty and secondary diagnostics

The full BF16 matrix is summarized primarily with five-fold means and fold dispersion. A separate 10,000-replicate, gold-stratified prompt-cluster bootstrap is preserved for the historical paired location comparisons of the original primary analysis. That bootstrap uses exact rendered-prompt clusters, applies the same multiplicities to both paired locations, and is conditional on fixed probes, selected layers, partitions, and representations. It supplies conditional paired intervals for the historical comparison; the complete 11-model BF16 matrix is summarized separately with fold-level results. Fold standard deviations retain their separate role as fold-dispersion summaries.

Fragmentation and error analysis are descriptive. Fragmentation is stratified by target token count in bins 1, 2, and 3+. The error artifact records per-class precision, recall, F1, support, confusion matrices, and recorded gold-POS strata. CKA and RSA are retained as fixed-sample diagnostics over 600 examples for the two primary models at the final layer. They serve as secondary diagnostics and remain outside layer selection, probe fitting, and the primary claims.

5 Prompt Conditions and Representation Interfaces

5.1 Frozen prompt templates

The prompt version is `bf16-lre-prompts-v1`; chat templates are disabled. The full-metadata template is:

```
Arabic morphology token probe.
Surface: {surface_dediac}
Token: {target}
Lemma: {lemma}
Root: {root}
Pattern: {abstract_pattern}
Predict the token morphology.
```

The metadata-free template is:

```
Arabic morphology token probe.
Surface: {surface_dediac}
Token: {target}
Predict the token morphology.
```

The rendered prompt, target text, character span, byte span, and prompt hash are stored per example. The templates are frozen before model comparison and remain unchanged during local analysis.

5.2 Information availability at each interface

The four interfaces are:

Full-metadata prompt-final (`full_prompt_final`) The final hidden state of the complete full-metadata prompt. The model can attend to the target, instruction, surface, lemma, root, and pattern fields, including fields after the target. These fields provide metadata that can correlate with morphology. This interface is therefore a contextual prompt endpoint; comparisons with target states combine position and information availability.

Metadata-free prompt-final (`metadata_free_prompt_final`) The final hidden state of the metadata-free prompt. It retains the surface and target but removes lemma, root, and pattern fields. It partially isolates the contribution of information availability while retaining a prompt-final context and the same task labels.

Target-final subtoken (`target_final_subtoken`) The state at the final tokenizer piece overlapping the target span. It is obtained from a causal prefix ending at that target-overlapping piece, excluding later prompt fields. It is derived from the complete target-span capture in the full-metadata bundle.

Target-mean span (`target_mean_span`) The arithmetic mean over all target-overlapping token states in the same causal prefix. It uses the complete target span and is derived locally after extraction using the existing forward pass.

The full-metadata prompt-final and target interfaces differ in both position and information availability. The metadata-free condition narrows this gap while retaining later instructional context; the target state remains prefix-only. We therefore describe interface comparisons as construct-validity analyses and report the conditions under which each location is evaluated.

5.3 Why retain the complete span

Saving the complete target span preserves later pooling choices. Each BF16 layer stores the concatenated hidden states for every token whose validated offset overlaps the target byte span. The local loader uses the recorded per-example token counts to derive the final row and the mean of the complete span. This supports final-subtoken, mean-span, and further target-span pooling analyses from the archived captures alone.

6 Models and Representation Extraction

6.1 Pinned model matrix

The matrix contains eleven official Hugging Face repositories: three Qwen models for a within-family size comparison, three Llama models for a within-family size comparison, Gemma, Phi, and Mistral as cross-family references, and Jais and ALLaM as Arabic-specialist references. The matrix is intentionally heterogeneous: it mixes base and instruction-tuned checkpoints and supports robustness across selected model families. Scale and training-data effects remain outside this design.

Every model has a frozen model revision, tokenizer revision, and config revision. These revisions are full commit SHAs in the machine-readable model manifest and are reproduced in Appendix A. The requested inference and saved tensor dtype is BF16. The manifest records native checkpoint dtype separately: Jais is listed with a native FP32 checkpoint and was loaded under the explicit BF16 extraction contract. The extraction contract excludes FP16 or FP32 fallback, quantization, CPU offload, and revision fallback.

6.2 Extraction semantics

The extractor uses the pinned Hugging Face revision and tokenizer, evaluation mode, a single CUDA device, right padding, and no chat template. It captures the embedding output and each transformer-block output before final model normalization. Logits and final-normalized representations are separate quantities outside these capture sites. Saved tensors are BF16. The capture sites are embedding output, prompt-final state per layer, and all target-span token states per layer; full sequence-by-full-layer activations are omitted.

At extraction time, the harness fails on any mismatch in access, revision resolution, tokenizer identity, architecture, context length, or BF16 requirements relative to the manifest. The exact prompt and model metadata are stored with each bundle. The completed analysis uses 22 validated model/condition bundles, two prompt conditions per model, and four analysis units per model.

6.3 Structural alignment and example identity

The target is rendered from a designated field and its character and UTF-8 byte spans are recorded structurally. A fast tokenizer with offset mapping must identify a contiguous complete overlap. The harness records token IDs, sequence length, token pieces, target token indices, target token count, the decoded target text, and the selected final overlapping token. It selects representations only from this validated mapping; assumed position, row order, `[-1]`, and unvalidated text reconstruction are excluded. Alignment and example identifiers must remain in the same order, and any failure prevents successful completion.

Each of the 22 bundles contains 4,701 examples and 4,701 successful alignments. The analysis audit therefore enters local probing only after bundle checks, metadata order checks, tensor shape checks, and finite-value checks pass. The complete analysis has 44/44 atomic model-by-interface units, 1,980 real result cells, and 360 shuffled result cells. All eleven models use five outer folds for random, lemma-heldout, and root-heldout evaluation.

6.4 Bundle and provenance schema

The bundle schema is versioned as `bf16-hidden-state-bundle-v1`. Each bundle records model and tokenizer identity, prompt condition and template, dataset path and hash, example metadata, alignment records, representation semantics, layer count, hidden size, device, requested and saved dtype, code identity, command, and runtime metadata. Scientific files are covered by a SHA-256 manifest. Completion is a separate atomic marker bound to the current checksum manifest and a verified persistent destination. A partial bundle remains ineligible for the successful marker.

This design matters scientifically because it separates model inference from analysis. The local sufficiency audit confirms that the saved fields support the split reconstruction, all planned pooling choices, layer curves, linear probes, majority controls, shuffled controls, fragmentation bins, class/POS error analysis, BF16/Q8 metric comparison, and CKA/RSA-compatible matrices. The recorded post-capture contract is `weights_required=false`.

7 Results

7.1 Overview of the complete matrix

The primary BF16 analysis contains 11 models, 4 representation interfaces, 3 tasks, and 3 split conditions, with 5 outer folds for every combination. It therefore yields 396 summary rows (11 x 4 x 3 x 3) and 132 rows in the compact random-minus-grouped macro-F1 comparison. The compact comparison defines

$$\Delta_{\text{lemma}} = F1_{\text{random}} - F1_{\text{lemma-heldout}}, \qquad \Delta_{\text{root}} = F1_{\text{random}} - F1_{\text{root-heldout}}.$$

Every one of the 132 rows has positive values for both deltas. Across rows, Δ_{lemma} ranges from +0.0127 to +0.2863 with mean +0.1059, and Δ_{root} ranges from +0.0255 to +0.2796 with mean +0.1179. These descriptive averages summarize five-fold means. The result establishes a consistent direction in this resource and matrix, while the range shows why a single universal "leakage penalty" would obscure the observed variation.

Table 2 shows the heterogeneity. The model-level mean lemma gap ranges from +0.0637 for ALLaM-7B to +0.1881 for Llama-3.2-1B; the root gap ranges from +0.0750 for Qwen3-8B to +0.2054 for Llama-3.2-1B. These grouped means provide descriptive orientation. Model ordering remains confounded by families, parameter counts, native training mixtures, instruction status, tokenizers, and architecture.

7.2 Effect of lexical split design

The complete result is stronger than a two-model demonstration. For every model, task, and interface, random-split macro-F1 is above the corresponding lemma-heldout and root-heldout value. The positive difference is consistent with random splitting

Table 2 Descriptive mean macro-F1 gaps by model. Each value averages the 12 task-by-interface rows for that model in the generated 132-row table. The full model-by-task-by-interface table is archived as `results/macro_f1_split_deltas.csv`, Markdown, and TeX.

Model	Random–lemma	Random–root
ALLaM-7B	+0.0637	+0.0816
Gemma-4-E2B	+0.1382	+0.1518
Jais-13B	+0.0711	+0.0918
Llama-3.1-8B	+0.0835	+0.0895
Llama-3.2-1B	+0.1881	+0.2054
Llama-3.2-3B	+0.1276	+0.1279
Mistral-7B	+0.0993	+0.1117
Phi-3-mini	+0.1036	+0.1314
Qwen2.5-1.5B	+0.1134	+0.1188
Qwen3-0.6B	+0.0977	+0.1118
Qwen3-8B	+0.0783	+0.0750

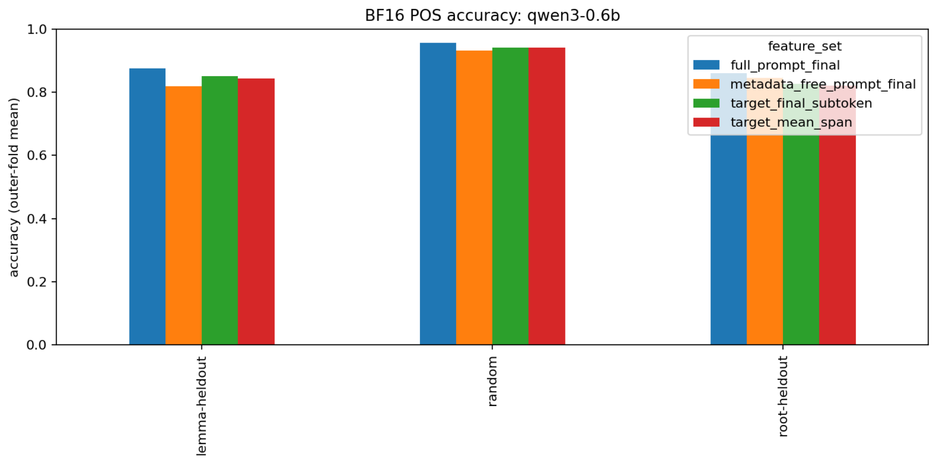

Fig. 1 Generated BF16 POS accuracy summary for Qwen3-0.6B across the three split conditions and four interfaces. The figure is descriptive; the full numeric matrix is the CSV artifact

allowing lexical identities or lexical families to be shared across training and test. The grouped conditions measure a more demanding form of transfer and leave open-set language generalization for separate evaluation.

The gap varies across grouping keys. In the full table, root-heldout gaps are larger on average than lemma-heldout gaps (+0.1179 versus +0.1059), but the model-level ordering differs: Qwen3-8B has a slightly larger lemma than root average, whereas most models have a larger root average. This variation is a reason to report both controls and keep their grouping keys visible.

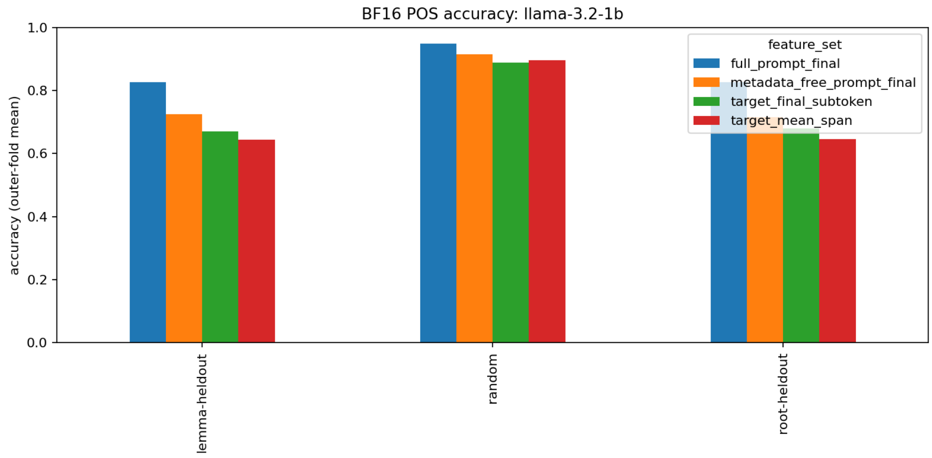

Fig. 2 Generated BF16 POS accuracy summary for Llama-3.2-1B. The paired figures illustrate model heterogeneity; causal architectural interpretation requires a controlled design

7.3 Heterogeneity across models and tasks

The interface/task summaries in Table 3 show that the largest descriptive gaps occur for POS when averaged over the matrix (+0.1339 lemma and +0.1498 root). Gender has smaller mean gaps (+0.0854 and +0.1009), and number is intermediate (+0.0982 and +0.1030). These descriptive values reflect this dataset, these prompts, these interfaces, and the fixed linear probe; they describe this matrix, with POS interpretation conditioned on the same evaluation choices.

Llama-3.2-1B is an informative high-sensitivity case. With target-mean-span pooling, its number gap is +0.2863 for random minus lemma-heldout and +0.2687 for random minus root-heldout; its POS gaps under the same pooling are +0.2785 and +0.2796. At the other end, the Mistral-7B full-metadata gender row has the minimum lemma gap in the generated table, +0.0127. These examples characterize heterogeneity: the amount of apparent random-split advantage depends materially on the model and task. The matrix supplies no controlled test of scaling or causal mechanisms.

7.4 Prompt metadata ablation

The full-metadata prompt-final state combines representation position with prompt information. In the Qwen3-0.6B lemma-heldout POS condition, full-metadata prompt-final macro-F1 is 0.8565, while metadata-free prompt-final macro-F1 is 0.7855. The corresponding target-final-subtoken and target-mean-span values are 0.8213 and 0.8116. This pattern is consistent with explicit metadata contributing to the information available at the contextual endpoint. The aggregate comparison leaves the responsible field or mechanism unresolved.

The ablation yields interface-specific effects. In the Jais-13B lemma-heldout number condition, metadata-free prompt-final macro-F1 is 0.8272 and full-metadata prompt-final macro-F1 is 0.7249. Similar reversals occur elsewhere in the full matrix.

Table 3 Descriptive mean macro-F1 gaps by interface and task. Interface rows average over models and tasks; task rows average over models and interfaces. All values are computed from the generated split-delta CSV.

Grouping	Random–lemma	Random–root
Full-metadata prompt-final	+0.0734	+0.0948
Metadata-free prompt-final	+0.1138	+0.1251
Target-final subtoken	+0.1141	+0.1179
Target mean span	+0.1222	+0.1337
POS	+0.1339	+0.1498
Gender	+0.0854	+0.1009
Number	+0.0982	+0.1030

Prompt-final interface comparisons must therefore be reported as information-availability comparisons as well as position comparisons. Importantly, metadata-free prompt-final and target interfaces remain above their shuffled controls in many settings, so the explicit-field ablation leaves substantial recoverable signal.

7.5 Target pooling and representation interface

Final-subtoken and mean-span pooling change absolute macro-F1 in some cells, but the larger split-dependent conclusion remains: all 132 interface-specific random-minus-grouped deltas are positive. Averaged over the matrix, target-final-subtoken has lemma/root gaps of +0.1141/+0.1179, while target-mean-span has +0.1222/+0.1337. The mean-span interface therefore shows somewhat larger descriptive split gaps in this matrix. Pooling order and linguistic interpretation remain conditional on this matrix and interface.

The complete-span capture is the important methodological safeguard. Because the raw states for every target-overlapping token are preserved, final-token and mean-span results are local transformations of the same evidence. A future pooling analysis within the recorded span is available from the existing forward pass. Capturing only a final subtoken would have precluded this robustness question.

7.6 Controls and class-sensitive interpretation

The 360 shuffled cells remain at or below their corresponding train-only majority accuracies. This is a necessary pipeline sanity result; causal use of the real-label results by the models requires separate evidence. Majority accuracy is particularly important for number. Across the matrix, the majority accuracy for number is approximately 0.8415 in the recorded summaries, while gender and POS majority accuracies are approximately 0.6261 and 0.6290. The number accuracy values can therefore look strong while macro-F1 remains limited for minority classes. All main interpretations use both metrics.

The per-class/POS artifact preserves the details needed to check this claim from archived evidence alone. It reports precision, recall, F1, support, and confusion matrices for each class, split, interface, and outer fold, together with gold-POS strata. The fragmentation artifact similarly reports accuracy and macro-F1 in the 1, 2, and 3+ target-token bins. These analyses are diagnostic descriptions of where errors occur, separate from the primary estimands.

7.7 BF16 relative to historical Q8

The historical Q8 captures provide robustness and historical context; BF16 is the primary experiment. For overlapping model/interface/task/split comparisons, BF16 and historical Q8 agree in majority-relative metric direction for 105 of 108 fold-zero comparisons. This supports the conclusion that the broad qualitative contrast between real-label results and the majority reference extends across two numeric formats.

The comparison measures metric-direction agreement; numerical equivalence between BF16 and Q8 hidden states would require paired hidden-state arrays. The raw historical Q8 feature arrays listed in the v3 external artifact record are absent from the live checkout, so direct BF16–Q8 vector cosine and relative-L2 comparisons remain unavailable. The historical Q8 run was neither repeated nor relabeled, and no vector claim is inferred from the metric-direction result.

7.8 Supporting CKA/RSA diagnostics

Fixed-sample CKA and RSA matrices are retained for the two primary models and are computed only as diagnostics. They document interface similarity while the task-level probe comparisons measure label recoverability; the two diagnostic families answer different questions. Since the split, ablation, and pooling results already answer the planned methodological questions, CKA/RSA remains a supporting result.

8 Discussion

8.1 What random and grouped splits measure

Random instance splitting measures performance when the probe may interpolate among instances drawn from the same lexical inventory as training. This can be a useful operational condition when deployment also contains repeated lexical identities; its optimism is a property of the estimand, and it remains a valid condition. Lemma-heldout evaluation measures transfer across lemma groups, and root-heldout evaluation measures transfer across root groups. The consistent positive gaps in this study show that these estimands differ in the Arabic resource. The appropriate condition depends on whether the claim concerns instance-level interpolation or transfer across lexical groups.

The varying size of the gaps matters alongside their direction. The heterogeneous gaps call for model-, task-, interface-, and grouping-specific reporting; a single penalty would overcorrect some combinations and undercorrect others. The compact table serves as an audit summary; any correction must remain specific to the model, task, interface, and grouping key. Researchers should report the random reference and the grouping key together, and should state whether the scientific claim concerns seen-lexicon interpolation or transfer across the grouping identity.

8.2 Why split sensitivity varies

Several mechanisms can produce the observed heterogeneity while leaving internal model strategy unresolved. Labels may be more lexically predictable for one task than another; a model may represent orthographic or contextual cues differently; tokenizers may expose different target fragments; and a prompt endpoint may integrate fields that are unavailable at the target. The matrix also confounds architecture, pretraining data, instruction tuning, parameter count, and tokenizer. The observed ordering therefore reflects a mixture of these factors, with causal attribution remaining underdetermined.

The matrix offers no clean size trend. The Qwen and Llama subsets provide within-family size comparisons, while the full design mixes architecture, training, instruction status, and tokenizer. The model-level table shows heterogeneity; claims about size-dependent lexical robustness require a more controlled design.

8.3 Representation-interface validity

The prompt metadata ablation clarifies a common ambiguity. A full-metadata prompt-final state may be highly predictive because the endpoint has access to fields that summarize lexical and morphological structure. That is a valid measurement if the question concerns the model's response to the complete prompt, while a target word state's recoverability requires its own comparison. The metadata-free condition narrows the information difference, while the target-prefix condition removes later fields. Both remain contextual and tokenization-dependent measurements.

Pooling is similarly part of the construct. The final target subtoken gives a specific causal position but may privilege the last fragment. Mean-span pooling uses all target-overlapping states and can change both signal and error patterns. Because substantive split conclusions survive both rules in the generated table, the lexical-split result survives the historical final-subtoken choice. That robustness is conditional on the captured span; generalization to other pooling schemes requires measurement.

8.4 Scope of Probe-Based Evidence

Probe performance characterizes information accessible to the estimator at the chosen representation under a named prompt, split, layer-selection rule, and estimator. The relationship between that accessibility and the model's causal generation mechanism remains outside the scope of this study. Lexical memorization, lexical generalization, linear recoverability, and linguistic competence describe different properties. Causal interventions, nonlinear estimators, and behavioral tests would answer different questions and belong to separate studies.

8.5 Methodological implications

For decoder-only morphology probing, the minimum reportable unit should include the lexical grouping rule, the prompt information available at the extraction site, the target-to-token alignment, the pooling operation, the metric and majority reference, and the boundary between development and test decisions. The present resource makes those decisions auditable and stores the representation evidence once. This permits later probes, split reconstructions, error analyses, and representation comparisons to run locally from archived evidence, eliminating repeated model inference and dependence on the survival of a cloud instance.

9 Limitations

First, the testbed is one resource: Modern Standard Arabic newswire derived from Arabic-PADT. The findings concern this resource; behavior for other Arabic genres, dialects, languages, or annotation traditions requires separate evaluation. The recovered source content matches official UD r2.17 and r2.18 training files, while the historical local release label remains unresolved because the original copy retained no Git metadata. The frozen dataset bytes and exact rebuild remain intact; the provenance ambiguity should remain visible.

Second, the tasks are POS, gender, and number. Arabic morphology includes additional features and interactions. The analyzer-derived lemma, root, and pattern fields are retained for grouping and metadata, while high-cardinality heldout-label classification falls outside the primary claim. Root and lemma are linguistically useful grouping variables with limited coverage of lexical identity.

Third, the primary estimator is linear and fixed. A nonlinear probe, a different regularization rule, or a different layer-selection policy would measure a different recoverability profile. The fixed alpha and development-only layer selection make the current comparisons auditable; estimator dependence remains.

Fourth, representation interfaces remain contextual and tokenizer-dependent. Metadata-free prompt-final still includes prompt context, and target states depend on model-specific subword boundaries. The ablation and pooling checks reduce known confounds; a unique "morphology location" remains unassigned.

Fifth, lexical grouping reduces direct recurrence. Open-set competence remains untested because surface similarity, morphological patterns, semantics, source regularities, and tokenizer fragments can cross heldout groups. Conversely, the model matrix mixes base and instruction-tuned models, so cross-model causal interpretation of these differences remains underdetermined.

Sixth, uncertainty summaries are conditional. The five-fold standard deviations summarize the recorded outer folds; the preserved prompt-cluster bootstrap targets conditional finite-test-set variation under fixed probes, partitions, and representations. Population or training-seed uncertainty and causal interventions require separate designs.

Finally, the direct BF16–Q8 vector comparison remains unavailable because historical Q8 feature arrays are absent from the live checkout. We report only the verified 105/108 majority-relative direction agreement. This limitation belongs to the historical artifact record; the Q8 experiment remains the historical execution with its original labels.

10 Resource, Code, and Reproducibility Availability

The complete working package is organized around immutable evidence and weight-free analysis. It contains:

- the frozen normalized stimuli, source release record, transformation code, field lineage, and dataset hashes;
- prompt templates, per-example rendered prompts and hashes, exact target character and byte spans, token IDs and pieces, and alignment records;
- the pinned eleven-model manifest with repository, revision, config, tokenizer revision, architecture, dtype contract, licensing/access status, expected dimensions, and resource estimates;
- validated BF16 hidden-state bundles containing all layer-wise prompt-final states and all target-span token states, together with manifests, commands, runtime metadata, checksums, and completion records;
- deterministic split manifests, five-fold result cells, primary-model predictions, majority and shuffled controls, fragmentation and class/POS error analyses, BF16/Q8 comparison records, and fixed CKA/RSA diagnostics;
- the local loaders and analysis runner, a sufficiency audit showing that planned post-extraction analyses can use archived representations alone, and a content-addressed BF16 analysis freeze.

The primary analysis source is `bf16-analysis-20260826-v2`. Its complete dataset and model-matrix hashes are recorded in the run manifest, frozen snapshot, and quantitative claim ledger. The freeze verifier passed for `evidence/bf16-analysis-20260826-v2/`. The historical LRE v3 freeze and historical Q8 run remain separate, immutable records; this manuscript leaves them unchanged.

Reproduction begins by verifying the dataset, model-matrix, bundle, and analysis-freeze hashes, then running the local analysis command against the frozen run root. Model weights are needed only for a new extraction. The current planned analysis state has `weights_required=false`; the saved captures are the scientific representation evidence.

11 Conclusion

This study evaluates lexical split design and representation-interface design as part of the measurement construct in decoder-only morphology probing. In the completed BF16 Arabic matrix, random instance splits produce higher macro-F1 than both lemma-heldout and root-heldout evaluation in every model–task–interface row, but the size of that advantage varies materially. The result supports a condition-specific interpretation of split gaps: split choice changes the question being answered.

The same qualification applies to representation position. A full-metadata prompt-final state can access explicit lemma, root, and pattern fields, while target and metadata-free interfaces make different information available and use different contextual boundaries. Final-subtoken and mean-span pooling provide locally recoverable alternatives when the complete target span is archived. Robust morphology probing should report these choices together and interpret scores as linear recoverability under the stated interface. Causal model use and unrestricted linguistic competence require separate evidence.

A Exact model identities and access metadata

Table 4 records the full model identity used by the BF16 run. The exact manifest additionally records expected BF16 weight size, VRAM estimate, context length, minimum and recommended AWS instance, gated status, access status, and trust-remote-code policy. The model list serves robustness across selected families, with scaling effects outside its design.

Table 4: Pinned BF16 model matrix. Repository revisions are full commit SHAs; tokenizer and config revisions equal the listed model revision for every row. The manifest is authoritative for licenses, gated status, access status, and all resource estimates.

Model	Architecture and size	HF repository and pinned revision	Dtype; hidden; role
`qwen3-0.6b`	`Qwen3ForCausalLM`; 0.6B	`Qwen/Qwen3-0.6B` `c1899de289a04d12100db370d81485cdf75e47ca`	bf16 / bf16; 1024 / 28; Qwen size-small
`qwen2.5-1.5b`	`Qwen2ForCausalLM`; 1.5B	`Qwen/Qwen2.5-1.5B-Instruct` `989aa7980e4cf806f80c7fef2b1adb7bc71aa306`	bf16 / bf16; 1536 / 28; Qwen size-medium
`qwen3-8b`	`Qwen3ForCausalLM`; 8B	`Qwen/Qwen3-8B` `b968826d9c46dd6066d109eabc6255188de91218`	bf16 / bf16; 4096 / 36; Qwen size-large
`llama-3.2-1b`	`LlamaForCausalLM`; 1B	`meta-llama/Llama-3.2-1B-Instruct` `9213176726f574b556790deb65791e0c5aa438b6`	bf16 / bf16; 2048 / 16; Llama size-small
`llama-3.2-3b`	`LlamaForCausalLM`; 3B	`meta-llama/Llama-3.2-3B-Instruct` `0cb88a4f764b7a12671c53f0838cd831a0843b95`	bf16 / bf16; 3072 / 28; Llama size-medium
`llama-3.1-8b`	`LlamaForCausalLM`; 8B	`meta-llama/Meta-Llama-3.1-8B-Instruct` `0e9e39f249a16976918f6564b8830bc894c89659`	bf16 / bf16; 4096 / 32; Llama size-large

Model	Architecture and size	HF repository and pinned revision	Dtype; hidden; role
gemma-4-2b	Gemma4For Conditional Generation; E2B	google/gemma-4-E2B-it 3e22461f65 e89153144f 8adb70e3b8 c2cc9845a7	bf16 / bf16; 1536 / 35; Cross-family
phi-3-mini	Phi3For CausalLM; 3.8B	microsoft/ Phi-3-mini-4k-instruct f39ac1d28e 925b323eae 81227eaba4 464caced4e	bf16 / bf16; 3072 / 32; Cross-family
mistral-7b	MistralFor CausalLM; 7B	mistralai/ Mistral-7B-Instruct-v0.3 c170c708c4 1dac9275d1 5a8fff4eca 08d52bab71	bf16 / bf16; 4096 / 32; Cross-family
jais-13b	JAISLMHead Model; 13B	inceptionai/Jais-13B 0d5eb87f91 e6577dac17 bc319eee94 0fb875b899	fp32 / bf16; 5120 / 40; Arabic-specialist
allam-7b	LlamaFor CausalLM; 7B	ALLaM-AI/ ALLaM-7B-Instruct-preview a28dd1e674 20cde72d36 29c8633a97 4cf7d9c366	bf16 / bf16; 4096 / 32; Arabic-specialist

The manifest's access metadata records Qwen3-0.6B, Qwen2.5-1.5B, Qwen3-8B, Gemma-4-E2B-it, Phi-3-mini, Mistral-7B-Instruct-v0.3, and ALLaM-7B as non-gated public entries; the Qwen entries use Apache-2.0, Gemma uses its terms of use, Phi uses MIT, Mistral uses Apache-2.0, and ALLaM's model license is recorded as a model-specific license requiring card verification. Llama 3.2-1B, Llama 3.2-3B, Llama 3.1-8B, and Jais-13B are gated entries with the respective Llama community or Jais access requirements in the manifest. Only Jais has `trust_remote_code=true`; all other rows use the manifest's false value. The tokenizer and config revisions equal the model revision in every row. These fields record model access and licensing directly in the frozen identity manifest.

B Representation bundle schema

For each model and prompt condition, the self-contained bundle has the following logical structure:

```
manifest.json
model.json / tokenizer.json / prompt.json
examples.jsonl
alignment.jsonl
representations/
```

```
  embedding_output.safetensors
  layer_0000.safetensors ... layer_<last>.safetensors
validation.json
checksums.sha256
COMPLETE
```

Each layer file contains a prompt-final matrix with shape $[N, H]$ and a flattened target-span matrix with shape $[\sum_i T_i, H]$, where N is the number of examples, H is hidden size, and T_i is the validated target token count for example i. The embedding file contains the embedding-output matrix with shape $[N, H]$. The complete target span makes final-subtoken and mean-span pooling local operations. Tensor validation rejects missing layers, shape mismatches, duplicate IDs, NaNs, and infinities.

C Complete result and audit artifacts

The complete model-by-task-by-interface split table is generated at `evidence/bf16-analysis-20260826-v2/snapshot/analysis/results/` `/macro_f1_split_deltas.csv`, with synchronized Markdown and TeX renderings. The source `summary.csv` contains 396 rows, each with five folds, accuracy, macro-F1, train-only majority accuracy, above-majority gap, and selected-layer records. The detailed cell JSON files preserve each fold's development layer metrics, selected layer, test metrics, and control metadata.

The audit package also contains:

- `shuffled_controls.csv` for the 360 primary-model control cells;
- `fragmentation.csv` with target token-count bins 1, 2, and 3+;
- `class_pos_error_analysis.csv` with per-class and POS-stratified errors;
- `bf16_q8_comparison.json` with the 105/108 majority-relative direction record and the unavailable vector-comparison status;
- `cka_rsa_diagnostics.json` and `analysis_decision.md`, documenting the supporting-only status of CKA/RSA;
- `local_analysis_sufficiency.json`, which records the fields consumed by the planned analysis and `weights_required_after_capture=false`.

D Quantitative claim and provenance ledger

The manuscript's claim-to-artifact mapping is supplied alongside this source as `quantitative_claim_map.md`. It maps every numerical claim to the generated result, dataset provenance, model manifest, or audit record that produces it. The manuscript reports numerical results only when a generating artifact is available; direct Q8 vector comparisons and ungenerated analyses therefore remain unquantified.

Acknowledgements. During preparation of this manuscript, the author used generative AI tools, including OpenAI Codex and DeepSeek-based tools, for limited editorial assistance and software/documentation support. The author reviewed and verified the manuscript, citations, code, data, analyses, and reported claims and retains full responsibility for the final work.

Statements and Declarations.

Funding. This study received no external funding. The work was self-funded by the author.

Competing interests. The author declares no competing financial or non-financial interests related to this work.

Ethics approval and consent to participate. Ethics approval was not required because this study analyzes a public, pre-existing annotated linguistic resource and involved no recruitment or interaction with human participants.

Consent for publication. Not applicable. The study uses a public, pre-existing linguistic resource and reports aggregate analysis results; no individual participant data are presented.

Data availability. The compact code, provenance, analysis, and validation package is publicly available in the source archive at https://github.com/voidwest/arabic-morphology-probing-lre. The repository contains the reproducibility manifests, generated results, validation reports, and checksums. The raw representation bundles, including those derived from Meta Llama, Jais, and ALLaM checkpoints, are not publicly redistributed. Any separate request is considered only after the requester independently obtains the corresponding checkpoint authorization, accepts the applicable model-card and license terms, and the transfer is permitted under those terms. No model weights, credentials, or Hugging Face tokens are shared; when these conditions cannot be confirmed, the raw bundle is not provided.

Materials availability. The dataset, prompt specifications, split manifests, model manifest, and derived analysis artifacts are described in the Resource, Code, and Reproducibility Availability section and in the public source archive at https://github.com/voidwest/arabic-morphology-probing-lre. Public redistribution is limited to the compact artifacts committed there. The license-aware policy for external raw bundles and card-restricted model derivatives is documented in `EXTERNAL_ARTIFACTS.md`; no raw gated-model bundle is anonymously downloadable from the archive.

Code availability. The extraction, validation, split, probe, and reporting code is included in the public source archive at https://github.com/voidwest/arabic-morphology-probing-lre.

Author contribution. Mohammed Al-Thobaiti: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing—original draft, Writing—review and editing, Visualization, and Project administration.

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